Coding & Solution

Date	24 September 2026
Team ID	SWTID-2026-2630
Project Name	AI Fit Track API
Maximum Marks	8 Marks

Coding & Solution – Implementation Snapshot

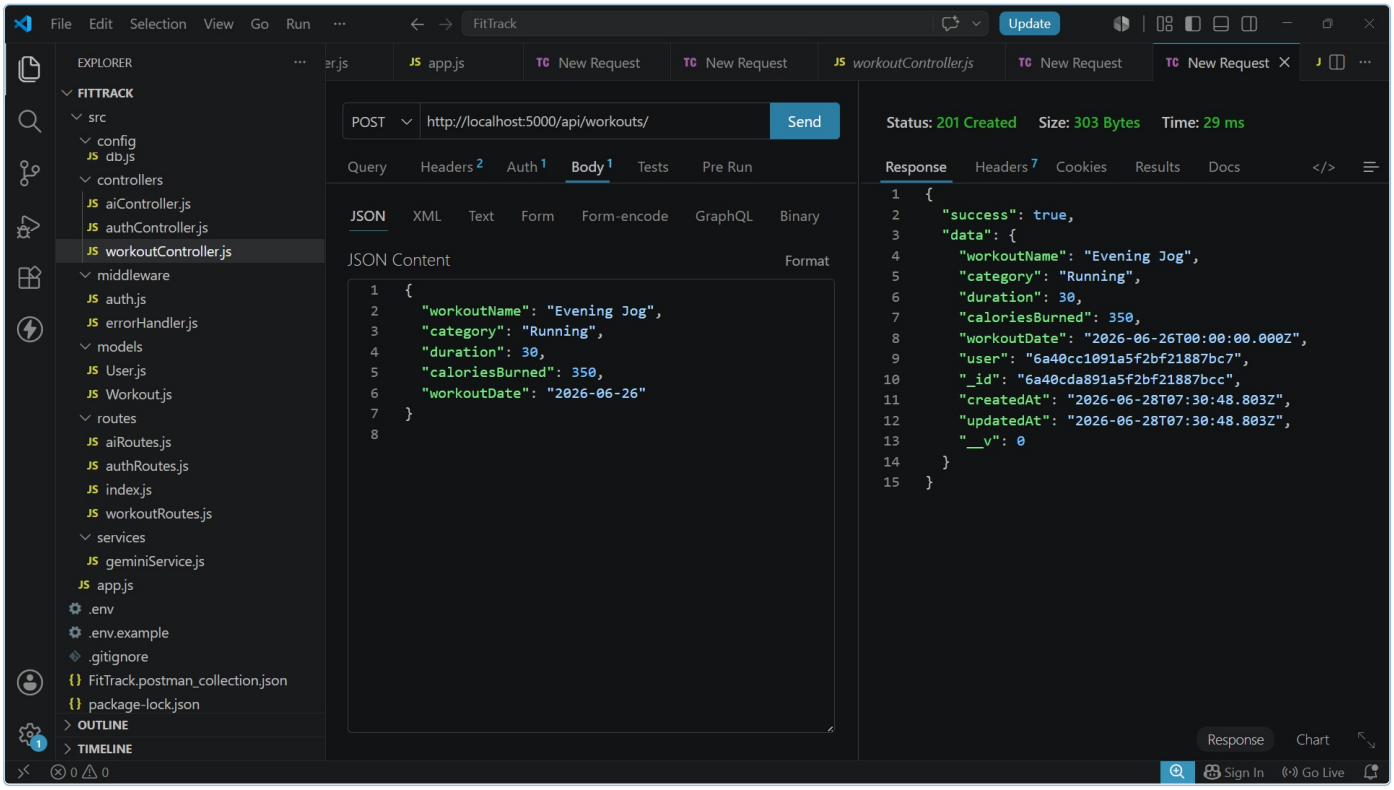


Fig. 1 – Actual project folder structure (captured mid-testing alongside a live API call)

Full source is version-controlled and available at https://github.com/thamaraiselvan6/fittrack_team_projt. The backend runs on Node.js/Express with MongoDB (Mongoose) and exposes Auth, Workout and AI route groups as described in the Solution Architecture.

Representative Endpoint (Add Workout)

```
POST /api/workouts/
Headers: Authorization: Bearer <JWT>
Body:
{
  "workoutName": "Morning Run",
  "category": "Running",
  "duration": 30,
  "caloriesBurned": 250,
  "workoutDate": "2026-09-20"
}

Response: 201 Created
{
  "_id": "...",
  "user": "...",
  "workoutName": "Morning Run",
  "category": "Running",
  "duration": 30,
  "caloriesBurned": 250,
  "workoutDate": "2026-09-20T00:00:00.000Z",
  "createdAt": "...",
  "updatedAt": "..."
}
```

Mongoose schema validation rejects the request before it ever reaches MongoDB if `category` is outside the fixed enum, `duration` is below 1, or `caloriesBurned` is negative – enforcing the Solution Requirements' business rules at the model layer.